Kyucheol Min

Cambridge, MA mink202@mit.edu kcmin02.github.io kcmin02 kyucheol-min-175187252
Google Scholar 0009-0002-3079-8845

Research Interests

- Generative modeling and neural sampling
- Machine learning for molecular and cellular systems
- Statistical mechanics

Education

Bachelor of Science (expected May 2027) — **Massachusetts Institute of Technology (MIT)**, Double major in Electrical Engineering and Computer Science (6-2) and Physics (8) — Cambridge, MA, May 2027

- Cumulative GPA: 5.0/5.0
- Highlighted coursework: Machine Learning (Grad), Deep Learning (Grad), Discrete Probability and Stochastic Processes (Grad), Algorithms for Inference (Grad, in progress), Quantum Computation (Grad), Electromagnetics (Grad), Quantum Physics II, Statistical Physics, Fundamentals of Statistics, Large-Scale Inference (in progress), Computer Architecture

Publications

FrOGS: Discrete Neural Sampler for Independent Alloy Configurations Across Chemical Conditions — 2026

K. Min, E. Hofgard, T. Smidt. arXiv:2609.02948. A multicomponent extension is in preparation for *npj Computational Materials*.
Kyucheol Min, Elyssa Hofgard, Tess Smidt
arxiv.org/abs/2609.02948 (NeurIPS AI4Mat Workshop (Spotlight))

Learning Lattice Parameters from Powder X-Ray Diffraction Data Using Invariants — 2026

E. Hofgard, **K. Min**, N. Segal, D. W. Mittan-Moreau, A. M. Tehrani, V. Oklejas, R. Gómez-Bombarelli, J. Nigam, D. W. Paley, A. S. Brewster, T. Smidt. arXiv:2607.21829.
Elyssa Hofgard, **Kyucheol Min**, Nofit Segal, David W. Mittan-Moreau, Aria Mansouri Tehrani, Vanessa Oklejas, Rafael Gómez-Bombarelli, Jigyasa Nigam, Daniel W. Paley, Aaron S. Brewster, Tess Smidt
arxiv.org/abs/2607.21829 (Acta Crystallographica Section A (accepted))

Experience

Atomic Architects Lab (PI: Tess Smidt), Undergraduate Researcher (Project Lead) — Mar 2026 – present, 9 months
FrOGS: Discrete Neural Sampler Across Chemical Conditions

- Conceived and developed a discrete neural sampler that couples an autoregressive model to a learned continuous-time Markov chain, generating independent configurations from the Boltzmann distribution across chemical conditions
- Matched exact free energies of the 2D Ising model on a 20×20 lattice to within 0.05% across its critical temperature
- Recovered all 3 CuAu ordered phases, whereas a published neural baseline missed $CuAu_3$, a thermodynamically delicate phase just below the convex hull of the free energy
- Used as a baseline by researchers at Microsoft Research and the University of Cambridge (He et al., arXiv:2610.01739)

Uhler Lab (PI: Caroline Uhler), Undergraduate Researcher (Project Lead) — July 2026 – present
Representation Learning of Single-Cell Images — 5 months

- Developed a self-supervised framework that separates perturbation-driven morphology from plate-to-plate technical variation; trained on 4.3M cells across 3,030 CRISPR gene knockouts
- Improved cross-plate replicate retrieval ~7× over DINOv2 and ~9× over Cell-DINO (mAP 0.0036 and 0.0027 → 0.0249) on held-out genes, and raised the share of knockouts above AUC 0.6 (22% vs 5–6% for the baselines)
- Confirmed the metrics were consistent on held-out genes from a function-disjoint split, suggesting generalization

Atomic Architects Lab (PI: Tess Smidt), Undergraduate Researcher — Sept 2024 – June 2026
Learning Lattice Parameters from Powder X-Ray Diffraction Data Using Invariants — 1 year 10 months

- Designed a transformer encoder architecture and trained it on the MP-20 dataset for bispectrum (an invariant lattice geometry descriptor) prediction from XRD patterns
- Showed that bispectrum intermediate representations outperform direct prediction for lattice parameter reconstruction from XRD data, reducing the mean absolute percentage error of cell lengths by ~2× on the experimental RRUFF dataset

Quantum Coherent Electronics Lab (PI: Kevin O'Brien), Undergraduate Researcher (Project Lead) — June 2025 – Sept 2026
Pulse Sequence Compiler for FPGA-Controlled Superconducting Qubit Experiments — 1 year 4 months

- Developed a pulse sequence compiler integrating Fermilab's Quantum Instrumentation Control Kit (QICK) into MIT's internal Python quantum framework (QCML), translating multi-channel experiment definitions and parameter sweeps into FPGA-native programs for qubit control and readout
- Demonstrated an end-to-end superconducting qubit characterization with the compiler, including T_1 and T_2 measurements
- Adopted by a team of 4 researchers at MIT Lincoln Lab for further development including multiplexing and active reset

Honors and Awards

54th International Chemistry Olympiad (IChO), Gold Medal — July 2022
Ranked 10th of 326 participants

Talent Award of Korea — Dec 2022
Awarded by the Korean Deputy Prime Minister and Minister of Education

Kwanjeong Scholarship — 2023
Full-tuition scholarship awarded to ~3 South Korean undergraduates per year (2023–present)

Eta Kappa Nu (HKN) — 2025
IEEE Honor Society for Electrical Engineering and Computer Science

Talks and Presentations

- Spotlight talk, NeurIPS AI4Mat Workshop (Dec 2026)
- Poster, Winter AI for Materials & Chemistry (AI4MC) Workshop (Dec 2026)

Teaching and Service

- Mentor of students in introductory electromagnetics (8.021)
- Co-founder of SIGMAS, the undergraduate statistics society

Skills

Languages
Python, C/C++, RISC-V Assembly

ML & Scientific Computing
PyTorch, pymatgen, e3nn, QuTiP, QICK

Tools
Git, Linux, SLURM